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Course/Code **:** DSA0308

Date **:** 10/07/26

QUESTION 1: REGULAR EXPRESSION PATTERN GENERATION

Code:

```
import re

text = """
Student Details:
Name: John Doe
Email: john.doe123@gmail.com
Mobile: +91-9876543210
Password: P@ssw0rd123
Date of Birth: 15/08/2004
Register Number: 23AIML1056
Department: Artificial Intelligence and Machine Learning
"""

# Regular Expression Patterns
email_pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
mobile_pattern = r'\+91-\d{10}'
password_pattern = r'[A-Za-z0-9@#$%^&*!]+'
dob_pattern = r'\d{2}/\d{2}/\d{4}'
register_pattern = r'\d{2}[A-Z]{4}\d{4}'

print("Email ID:")
print(re.findall(email_pattern, text))

print("\nMobile Number:")
print(re.findall(mobile_pattern, text))

print("\nPassword:")
password = re.search(r'Password:\s*(\S+)', text)
print(password.group(1))

print("\nDate of Birth:")
print(re.findall(dob_pattern, text))

print("\nRegister Number:")
print(re.findall(register_pattern, text))
```

Output:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Us
rs/ASHARAF/Desktop/DSA0308- NLP/Q1-AT1.PY"
Email ID:
['john.doe123@gmail.com']

Mobile Number:
['+91-9876543210']

Password:
P@ssw0rd123

Date of Birth:
['15/08/2004']

Register Number:
['23AIML1056']
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

QUESTION 2: SEARCH FOR A STRING

Code:

```
import re

paragraph = """
Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand.
"""

word = "AI"

match = re.search(word, paragraph)

if match:
    print("First occurrence found.")
    print("Starting Position :", match.start())
    print("Ending Position   :", match.end()-1)

    total = len(re.findall(word, paragraph))
    print("Total Occurrences :", total)

else:
    print("Word not found.")
```

Output:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q2-AT1.py"
First occurrence found.
Starting Position : 26
Ending Position   : 27
Total Occurrences : 6
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

SECTION 3: SPLIT THE DOCUMENTATION INTO SENTENCES AND WORDS

Code:

```
import re

paragraph = """
Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand.
"""

print("Original Paragraph:\n")
print(paragraph)

# Split into sentences
sentences = re.split(r'[.!?]+', paragraph)

print("\nSentences:")
count = 1
for sentence in sentences:
    sentence = sentence.strip()
    if sentence:
        print(f"{count}. {sentence}")
        count += 1

# Split into words
words = re.split(r'\s+', paragraph.strip())

print("\nWords:")
print(words)

print("\nTotal Sentences :", len([s for s in sentences if s.strip()]))
print("Total Words :", len(words))
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Use
rs/ASHARAF/Desktop/DSA0308- NLP/Q3-AT1.py"
Original Paragraph:


Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand.


Sentences:
1. Artificial Intelligence (AI) is transforming industries across the world
2. AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences
3. Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making
4. As AI continues to evolve, professionals with AI skills are in high demand

Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used
', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis,', 'banking', 'to', 'detect', 'fraud,', 'and', 'in',
 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'i
n', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', '
As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand.']

Total Sentences : 4
Total Words : 59
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

SECTION 4: EMAIL AND STRONG PASSWORD VALIDATION

Code:

```
import re

email = input("Enter Email ID: ")
password = input("Enter Password: ")

email_pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'

password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$%^&*!]).{8,}$'

# Email Validation
if re.fullmatch(email_pattern, email):
    print("\nEmail ID is Valid")
else:
    print("\nEmail ID is Invalid")

# Password Validation
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q4-AT1.py"
Enter Email ID: john.doe123@gmail.com
Enter Password: P@ssw0rd123

Email ID is Valid
Strong Password
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

SECTION 5: COMPILE ONCE, VALIDATE MANY

Code:

```
import re

emails = [
    "john@gmail.com",
    "student123@yahoo.in",
    "invalidmail.com",
    "admin@college.edu",
    "abc@xyz",
    "nlp_user@gmail.com"
]

pattern = re.compile(r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$')

print("Email Validation Result\n")

for email in emails:
    if pattern.fullmatch(email):
        print(email, "-> Valid")
    else:
        print(email, "-> Invalid")
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q5-AT1.py"
Email Validation Result

john@gmail.com -> Valid
student123@yahoo.in -> Valid
invalidmail.com -> Invalid
admin@college.edu -> Valid
abc@xyz -> Invalid
nlp_user@gmail.com -> Valid
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```